PHYS20352 Statistical Mechanics 2024-25: Notes on examination

This is the sixth year I have given this course, and the sixth year that the course has run in its current form. This document provides some notes on past and (possible) future exams to help with revision.

The course sections which contain material which is not examinable are

- Section 3.4 on information theory—but the Gibbs entropy, introduced in section 3.3, IS examinable
- Section 3.13 - specific problems using classical mechanics will not be asked
- Section 4.1.3 - details of specific heat derivation from appendix are not required—but the qualitative derivation is examinable
- Section 4.3.2 - details of the variable density derivation of the mass-radius relation, but the constant density energy minimisation derivation is examinable
- Section 4.4.3 - details of heat capacity above T_C.

I would also not set exam questions which only involved material from section 1.1-1.5 (other than definitions of free energies and their relation to observable quantities via partial derivatives, of course) but I hesitate to call it non-examinable because it is so important for the rest of the course.

As always, the quizzes and the examples sheets for the course provide the best guide to the examinable material, with the provisos that (i) tutorial sheets don't contain bookwork, (ii) tutorial questions are set in the expectation that you can consult your notes and past tutorial sheets and (iii) some examples are clearly too long to be appropriate for an exam.

The structure is the standard format with a multi-part compulsory question 1 and a choice of 2 from 3 other questions. There will be a formula sheet at the start of the exam with useful integrals and a couple of other things. It is attached at the end of this note. It is designed to be useful in *any* exam, so it is not a guide to what is actually on this year's exam.

In 2017/18 and for decades before, the material of this course was spread over two courses, PHYS20352 Thermal and Statistical Physics and PHYS30151 Bose and Fermi Gases. About half the material of PHYS20352 now resides in PHYS10352, and will only be examined in the current course in the context of statistical physics (no heat engines, for instance). So everything on old PHYS20352 exams should be familiar to you. There was quite a bit of repetition between the two courses, so in fact almost all of PHYS30151 is also covered in the current course and almost all questions are accessible. The exceptions are those on the properties of graphene, though as graphene can be treated as a 2D ultrarelativistic Fermi gas with zero chemical potential, many question parts even on graphene remain relevant.

The density of states may be denoted $\frac{\mathrm{d}n}{\mathrm{d}k}$, $D(k)$ or $g(k)$ (and similarly for $g(\varepsilon)$). It always includes the volume or area, so it is a density per unit k but not per unit volume. In that the usage differs from some other courses.

P.T.O

The following are lists of questions from past exams that I **would not** set (which isn't to say that ideas within them cannot be invoked). I compiled it when this course was new in 2020, perhaps it will still be of interest particularly as two of the years since then were covid-affected.

PHYS20352

Previous examiners of PHYS20352 have been fond of the classical calculation of the partition function (section 3.13), but it is not central to this course.

- 2019: 1b
- 2018: 1c; 2
- 2017 1b; 2; 3b,c
- 2016 1a,b,c; 2
- 2015 1b,c; 3b
- 2014 1b,c; 2, 3, 4

PHYS30151

Note that in 2016 and before, the lecturer set $k_B \to 1$, that is T has units of energy, and entropy and heat capacities are dimensionless. Restoring the missing factors of k_B is a good exercise! Questions about graphene would be phrased more helpfully.

- 2019 all OK
- 2018 all OK
- 2017 all OK
- 2016 2c
- 2015 1c
- 2014 2b,c

P.T.O. for formula sheet.

Formula sheet

Information which may be used in this paper:

The density of states for a spinless particle in two dimensions is $g(k) = \dfrac{Ak}{2\pi}$.
The symbol β is defined as $\beta = 1/(k_B T)$.

You may use $k_B = 8.6173 \times 10^{-5}$ eV K^{-1}

The symbol n_Q is defined as $n_Q = \left(\dfrac{mk_BT}{2\pi\hbar^2}\right)^{3/2}$.

The translational partition function for a single non-relativistic particle in a box is $Z_1 = V g_s n_Q$, where g_s is the spin degeneracy.

In the classical limit, the chemical potential for an ideal gas is given by $\mu = -k_BT \ln\left(\dfrac{g_s n_Q}{n}\right)$.

The Planck formula for the energy density of black-body radiation is

$$u(\omega) = \frac{\hbar}{\pi^2 c^3}\,\frac{\omega^3}{e^{\hbar\omega\beta} - 1}.$$

Stirling's approximation is $\ln N! \approx N \ln N - N$ for $N \gg 1$.

$\displaystyle\sum_{n=0}^{\infty} (n+1)\, x^n = (1-x)^{-2} =$ for $|x| < 1$.

The following integrals may be useful (note $n!! \equiv n(n-2)(n-4)\ldots 1$ for odd n.)

$$\int_0^\infty \frac{x^{1/2}}{e^x+1}dx = 0.678094 \qquad \int_0^\infty \frac{x^{1/2}}{e^x-1}dx = 2.31516$$

$$\int_0^\infty \frac{x}{e^x+1}dx = \frac{\pi^2}{12} \qquad \int_0^\infty \frac{x}{e^x-1}dx = \frac{\pi^2}{6}$$

$$\int_0^\infty \frac{x^{3/2}}{e^x+1}dx = 1.15280 \qquad \int_0^\infty \frac{x^{3/2}}{e^x-1}dx = 1.78329$$

$$\int_0^\infty \frac{x^2}{e^x+1}dx = 1.80309 \qquad \int_0^\infty \frac{x^2}{e^x-1}dx = 2.40411$$

$$\int_0^\infty \frac{x^{5/2}}{e^x+1}dx = 3.08259 \qquad \int_0^\infty \frac{x^{5/2}}{e^x-1}dx = 3.74453$$

$$\int_0^\infty \frac{x^3}{e^x+1}dx = \frac{7\pi^4}{120} \qquad \int_0^\infty \frac{x^3}{e^x-1}dx = \frac{\pi^4}{15}$$

$$\int_0^\infty x^n e^{-x/a}dx = n!\,a^{n+1} \qquad \int_0^\infty x^n e^{-x^2/a^2}dx = \frac{(n-1)!!\sqrt{\pi}a^{n+1}}{2^{(n/2+1)}} \quad \text{for even } n.$$